Computational thinking

1. **Explain abstraction with example**

-> Abstraction is the act of representing the essential features without including background details and explanation. It is also the collection of data and methods. An abstract data type is a programmer-defined data type that can be manipulated like system-defined data types.

For example:- in a “switch-board”, you only press certain switches according to your requirement. What is happening inside, how it is happening, you needn’t know. This is an Abstraction.

2. **Explain decomposition with example**

-> Decomposition means breaking a large, complex problem or system down into smaller, manageable, and solvable parts.

For example:- in “video game design”, Splitting a game development project into separate sub-tasks like storyline creation, character design, physics engine, scoring system, and user controls.

3. **Explain pattern recognition with example**

-> Pattern recognition is the process of looking for similarities, trends, or regularities among decomposed, smaller problems. Once you break a large problem down, pattern recognition helps you identify shared characteristics that allow you to solve those pieces more efficiently by reusing proven solutions.

For example:- Traffic lights are an example of pattern recognition.

The sequence Red → Yellow → Green repeats continuously. By recognizing this repeated pattern, we can predict which light will appear next.

4. **Explain algorithm design (first flow chart) with example**

-> Algorithm design is the process of developing step-by-step set of statement that solve a particular problem.

-> Flowchart is a diagrammatic representation that illustrates the sequence of operations to be performed. Designers use flowchart as a program planning tool for visually organizing a sequence of steps necessary to solve a problem.

For example:- find the sum of two numbers.

Step 1:- Start

Step 2:- Read two number A and B.

Step 3:- sum=A+B.

Step 4:- Display sum.

Step 5:- Stop

Web Development

Explain how internet works?

-> The internet works by connecting billions of devices worldwide into a massive network of smaller networks that share information using standard rules. When you request a website, the Domain Name System (DNS) acts like a phone book, translating the human readable web address into IP address so your computer knows exactly where to sent the request. The website's data is then divided into tiny pieces called packets, which travel dependently across various routers to find the fastest path in your device. Once all the packets arrive at their destination, your phone or computer instantly reconstitutes them in the correct order to display the page on screen.

Explain client server architecture with figure?

->Client-server architecture is a centralized computing model split into two main roles: the client, which requests data, and the server, which provides it.

This architecture is broadly used network model. In client server architecture, clients and servers are differentiated, specific server and clients are present. In client server architecture, centralized server is used to store the data because its management is centralized. In client server architecture, server responds to the services which are requested by clients

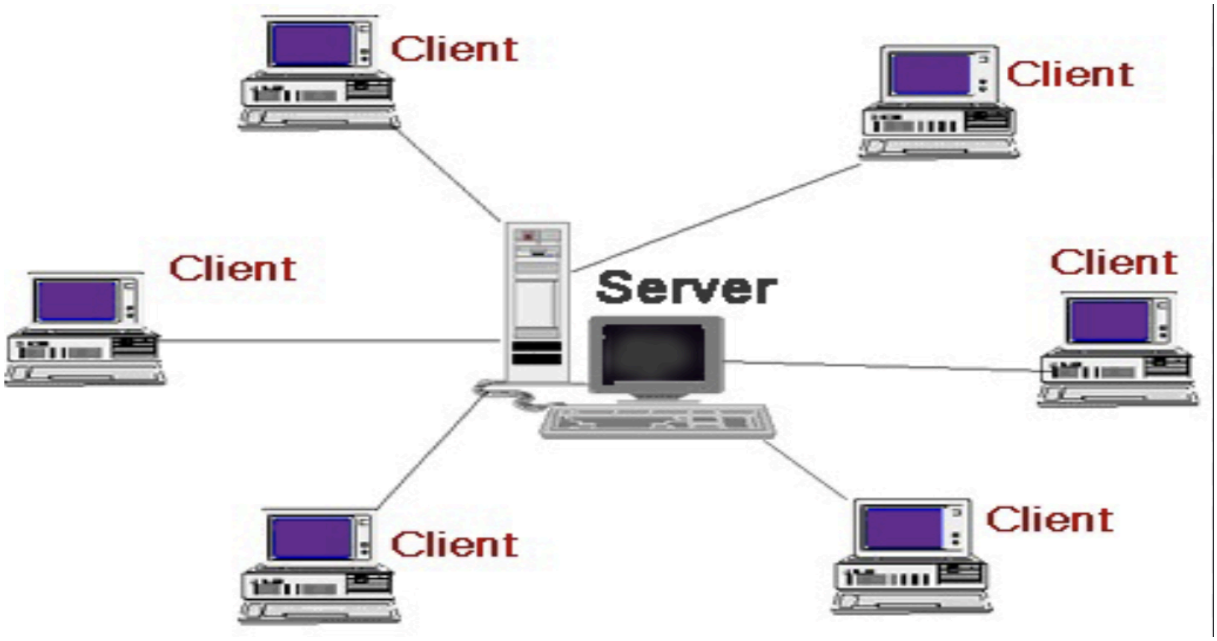


Fig: client-server architecture

Explain markup language vs programming language.

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Markup Language	Programming Language
A system for interpreting a document in a way that is grammatically distinguishable from the text.	A formal language that contains a set of instructions used to produce various kinds of output.
Used to present information.	Used to give instructions to a computer to perform a particular task.
Does not usually contain logic or algorithms.	Does not usually contain logic or algorithms.
Examples:- HTML	Example:- JAVA, Python, C++

Difference between http and https.

->

http	https
It is hyper text transfer protocol.	It is hyper text transfer protocol secure.
It is not secure.	It is more secure.
It uses Port 80 by default.	It uses Port 443 by default.
HTTP URLs begins with http://.	HTTPS URLs begins with https://.
Data is sent without encryption.	Data is encrypted using TLS(Transport Layer Security) / SSL(Secure Sockets Layer)

Explain html tags, opening tags and closing tags with example

-> **HTML tags** are the fundamental building blocks of a webpage. They tell the web browser how to structure and display content, such as text, images, and links. Most tags come in pairs: an opening tag (<tag>), the content, and a closing tag (</tag>).

An **opening tag** marks the start of an HTML element and tells the web browser where the effect of that element begins.

A **closing tag** is a piece of code that marks the end of an element in markup languages like HTML and XML.

Example:-

```
<html>
<head>
<title>My First Web Page</title>
</head>
<body>
<h1>Hello, World!</h1>
<p>This is a paragraph.</p>
</body>
</html>
```

Output:-



**Create a html Page and implement all these tags : <p>,
,<hr />,heading (h1-h6), <!--...→,<ol>, <ul>, <img />, <a>**

->

day1.html > html

```html
<!DOCTYPE html>
<html>
<head>
    <title>HTML Tags</title>
</head>
<body>

    <!-- This is a comment -->

    <h1>Heading 1</h1>
    <h2>Heading 2</h2>
    <h3>Heading 3</h3>
    <h4>Heading 4</h4>
    <h5>Heading 5</h5>
    <h6>Heading 6</h6>

    <p>This is a paragraph.</p>

    Hello<br />World

    <hr />

    <ol>
        <li>Apple</li>
        <li>Banana</li>
    </ol>

    <ul>
        <li>Red</li>
        <li>Blue</li>
    </ul>

    <img src="image.jpg" width="200" />

    <p>
        <a href="https://google.com">Google</a>
    </p>

</body>
</html>
```

